Scalability & Future Plan

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact & Operational Risk	Priority to Address
1	**Local SQLite Concurrency Bottleneck**	SQLite single-threaded write locks cause disk I/O queuing when multiple site supervisors upload audit telemetry simultaneously.	**High**
2	**Cloud Inference Quota & Latency**	Reliance on external Groq cloud LPUs requires stable uplink bandwidth; weak remote connectivity falls back to offline guidance without bounding boxes.	**Medium**
3	**Static Video/Photo Sampling**	The prototype currently processes discrete snapshots and scheduled intervals rather than 30 FPS continuous multi-feed RTSP streams.	**Medium**

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Architectural Solution
1	**User Load**	Handles ~50 concurrent supervisors on a single FastAPI instance.	Deploy horizontally auto-scaling FastAPI containers on Kubernetes (EKS) behind Nginx load balancers.
2	**Data Storage**	Local SQLite3 database file and local filesystem.	Migrate to Amazon Aurora PostgreSQL clustering and archive high-res drone imagery to encrypted AWS S3 buckets.
3	**Performance**	Groq Cloud Vision API calls with ~950 ms network roundtrip.	Deploy quantized edge models (YOLOv11 + TensorRT) directly on on-site NVIDIA Jetson Orin edge boxes.
4	**Security**	Session tokens, bcrypt password hashing, and local .env secrets.	Implement enterprise Single Sign-On (SSO / SAML 2.0), AWS Secrets Manager, and SOC-2 compliant end-to-end audit encryption.

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Future Roadmap (Phase 1 to Phase 4):

Long-term strategic roadmap detailing multi-phase architectural milestones, technical enhancements, target deployment timelines, and projected enterprise ROI across site safety and structural engineering operations.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact & Industry ROI
Phase 1 MVP (Current)	• **Multimodal PPE Detection:** Hard-hat, vest, and harness violation tagging. • **IS 456 Structural Crack Diagnostics:** Surface crack width classification and chemical repair blueprints. • **Weather & BOQ Risk Advisory:** Doppler rain radar sync and automated takeoff estimates. • **Automated PDF Engine:** Instant executive compliance reports with zero data retention.	**Q4 2026** **(Completed)**	• Eliminates 80% of manual safety inspection time. • Provides sub-second hazard feedback directly to site engineers. • Prevents concrete batch washouts during unexpected monsoons.
Phase 2 Edge & Autonomous	• **Autonomous Drone Integration:** Programmed waypoint flights for 360-degree aerial site scanning. • **On-Site Edge Box Deployment:** Quantized vision inference on local NVIDIA Jetson Orin modules. • **Zero-Latency Sirens:** Direct GPIO-triggered audio sirens when un-helmeted workers enter crane zones.	**Q1 – Q2 2027**	• Sub-200ms real-time audio alerts without relying on internet uplinks. • Full perimeter coverage over 50+ acre infrastructure project sites. • Completely eliminates site observer fatigue.
Phase 3 BIM & ERP Integration	• **Autodesk Revit / Navisworks Sync:** Automatically pins detected concrete cracks directly into 3D BIM structural models. • **Enterprise ERP Automation (SAP / Oracle):** Direct purchase requisitions generated for recommended epoxy resins and carbon-fiber wraps. • **Contractor Rating Portal:** Subcontractor safety scoring logs.	**Q3 – Q4 2027**	• One-click repair material procurement reducing maintenance lead time by 60%. • Seamless digital handover of building health records to client facilities management.
Phase 4 Predictive Digital Twin	• **Structural Digital Twin:** Multi-sensor IoT data fusion (vibration, strain gauges, ultrasonic sensors) combined with vision history. • **Predictive Degradation Models:** 10-year concrete crack propagation and seismic resilience simulations. • **Compliance Blockchain:** Tamper-proof municipal audit logs.	**Q1 – Q2 2028**	• Accurate lifecycle forecasting preventing premature structural demolition. • 20–30% reduction in contractor liability insurance premiums. • Benchmark certification for green and disaster-resilient building standards.